TREATMENT OF CHRONIC ALCOHOLISM WITH LYSERGIC ACID DIETHYLAMIDE*

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An increasing use is being made of Lysergic acid Diethylamide (LSD-25) in the treatment of psychoneurotics and chronic alcoholics, and the following is a further report on our findings with LSD-25 as an adjuvant to psychotherapy with chronic alcoholics. This is part of a research programme carried out in Saskatchewan, after Smith first gave a preliminary report on his findings with this type of therapy (7), which was followed up by a more extensive study (8) (9). A report on the use of LSD-25 in the treatment of alcoholism in British Columbia was published by MacLean (5). A review of the therapy as carried out in the Saskatchewan Hospital in Weyburn, has recently been published (3). It is felt that the experience gained in this Alcoholic Unit and the technique followed there, might be of some general interest and it is presented here, illustrated by short case histories.

The Treatment Programme

The treatment programme has consisted of two months' continuous hospitalization, during which period of time the patient was integrated into a group. He received a single experience with LSD-25 during the last week of his stay. For a full description of the technique and the career of the alcoholic in the hospital, see (3) (9). Considerable emphasis was laid on the fact that the therapy was directed along the lines and philosophy of Alcoholics Anonymous. This means, among other things, that religion and the concept of a "Higher Power" becomes a central theme in group-therapy as well as in the LSD experience.

The "tasteful physical setting", as described by MacLean *et al*, we find idealistic for a mental hospital. It is undoubtedly important for setting the tone at the start of the experience, but the rapport built up between patient and therapist is more important and can compensate for deficiencies in the physical surroundings.

The routine dosage has been 200 gamma. It is generally felt that higher dosages do not result in more therapeutic effects. The resulting confusion or amnesia is undesirable and sometimes even disturbing to the patient when later, he is trying to establish for himself the reality of the experience. An exception is illustrated in case 5.

The drug is administered at 9 a.m. and the junior therapist remains with the patient all day, while the senior therapist spends part of the day in the session. As most workers with LSD stress, the therapist must be familiar, through several personal experiences, with the effect of the drug, but we agree with MacLean, that it is not a good idea for the therapist to take the drug at the same time as the patient.

The Experience

The sequence of the experience has been discussed elsewhere (1) but each experience differs markedly from those experienced by other people, or by the same person on different occasions. We feel that it is necessary for the therapist to be well acquainted with the patient and have a good relationship with him prior to the session. This way, and using his knowledge of the effect of the drug, he will often be able to direct or channel the patient's thoughts during the session towards possible problem areas in the experience. This channelling can be done by means of music, pictures or an occasional question directed to the patient and should not be confused with a forced indoctrination of the therapist's values.

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LSD-25 has been described as a psychotomimetic (psychosis mimicking) agent and it is felt that some direction has to be given to the patient in order to emphasize the psychedelic (mind manifesting) aspects of the experience while deemphasizing such symptoms as disordered perceptions or feelings of unreality. This is one of the reasons why it is important for the therapist to be familiar with the drug reaction from personal experience.

Three aspects of the experience will be described, namely, the abreaction, the transcendental experience and the breaking down of defences, since it is felt that these are some of the more therapeutic factors involved in the experience.

1. *The Abreaction.* Some patients, usually quite early in the session when the drug is just taking effect, will relive a traumatic childhood experience which has been repressed. This is usually preceded by intense anxiety or possibly nausea and after the abreaction there is a definite release of emotional tension. We feel that what some writers have called the "side effects of the drug—anxiety, nausea, tension, headache and so on, are indicative of emotional conflicts and pass away when the emotional conflicts have been resolved. Indeed it is almost always possible to treat these symptoms as emotional defence mechanisms.

In some cases, where the therapist knows that a certain period in the patient's past is worth investigating, it can often be made part of the experience if it is made the subject of casual discussion during the half hour or so from the time the drug is taken until it takes effect. In other cases, repressed material is brought out spontaneously. The therapeutic value of such abreaction is perhaps difficult to assess, but it occurs on occasions with or without the help of the therapist and there is a noticeable diminuation of anxiety in the patient when he can view the experience in the light of his present adult knowledge. The following example will illustrate this:

Case 1. S.H.

The patient was a 30 year old single radar technician of English background, who was admitted to the hospital for the third time, although the two previous times he had received only so-called "drying up" treatment. Once he had discharged himself against medical advice.

He was a somewhat dull looking and slovenly man of mesomorphic build who, at the time of admission, felt that he drank in response to periods of frustration and anxiety and thought he had now come to the end of his rope, and realized that he could not continue like this. He appeared, however, to have very little real understanding or insight into of the nature of his problems, and he seemed to feel that his drinking was due to lack of will-power and consequently he would be able to stop drinking anytime he put his mind to it. During the course of his hospital stay the patient showed repeatedly that he possessed a very low frustration tolerance and also that he was unable to deal with frustrations in an adequate manner.

Early in the LSD experience he stated that he had a sore leg and added that he always felt this whenever he got drunk. He became more and more anxious and finally relived an incident which occurred when he was eight or nine years old. During a race on bicycles with other children, he fell and hurt his leg and was subsequently scolded by his father for having lost the race. It may be noted that this incident formed only a small part of the LSD session, most of which took the form of a psychotherapeutic discussion of his relationship with his mother and his wife. When last heard of six months after discharge, he had remained sober in spite of living under considerable marital disharmony.

2. *The transcendental experience.* This occurs usually from one to four hours after administration of the drug and only with doses over 100 or 150 gamma. The session can probably be controlled so that

the patient does not go through this experience, but we feel that it is a worthwhile stage and can have a profound influence on the patient. It is closely related to the surrender or conversion experience described by Tiebout. In the LSD experience there are at least two factors working towards the achievement of the "surrender". One is the surrender to the drug itself. As MacLean *et al* have pointed out, attempts by the patient to misuse or control the experience can produce extreme physical and psychological discomfort. The patient, therefore, finds himself in a situation that he cannot control or manipulate to suit himself; if he tries it becomes too painful, so that he has to submit himself to the experience. The second factor working towards this surrender is the transcendental experience. Many reports of LSD experiences allude to an out-of-this-world experience in which the subject has a feeling of being at one with God. This experience has, in many cases, made it easier for the patient to accept A.A.'s concept of a "higher power". However, the patient's reaction to his spiritual experience is, of course, intimately connected with what religion, through his past experiences, has come to mean to him. A transcendental experience with a spiritual flavour is such a strong and powerful emotional event that the patient is overwhelmed and he experiences what can be described as a conversion. This is not a conversion to a particular church or doctrine and need not, or should not, be a projection of the therapist's belief or faith, but rather a reaffirmation of the patient's own belief. A transcendental experience does not always occur but is facilitated if the session is allowed to develop in an atmosphere of pleasantness and acceptance. A spiritual experience does not occur in every transcendental experience but can often be brought out by the use of religious music and pictures.

With this experience and knowledge of a "higher power" comes the humility which Tiebout (12) agrees is so important in maintaining sobriety. It is a giving up of control that is forced upon the alcoholic in the LSD session. If he tries to maintain control during this transcendental or spiritual experience he suffers intense anxiety or physical discomfort, but once he lets go he finds he can be at peace with himself. One patient trying to describe this surrender said he had to lose, but once he had lost he found he had won. Compare this statement of an alcoholic with a quotation from Suzuki's discussion on '*Human Values in Zen*" (4), "It is strange that we have to lose in order to gain". Numerous reports concerning the achievement of sobriety following a conversion experience can be found and Osmond's original premise, (6) concerning the effectiveness of an overwhelming experience, appears to be brought out in our observations. This spiritual experience is not so much an intellectual knowledge of a "higher power" as it is an emotional feeling and with it comes self-understanding and self-acceptance. Many therapists report that the alcoholic is immature and dependant and Walberg (13) suggests that "Many alcoholics stop drinking when they feel that their dependency is appeased by an alliance with God". This alliance with God can be a strong emotional experience under LSD, as the following case will illustrate:

Case 2. S.H.

A 47 year old married man with a traumatic childhood history and a lifelong history of alcoholism and stealing, gave the following account of his LSD experience. "I found myself somewhat at a loss as to just how to describe, in a few words, what took place after some 47 years of beating my brain out against a wall of indifference, self-centredness, and ignorance, plus the inability to believe there could be a greater power than me."

"Today that wall was ripped apart, and with it, I found myself faced with the prospect of making the supreme choice,

I did not give up without a struggle, true to form as established in a pattern laid over a period of 47 years, I was reluctant to consider the existence of a power greater than myself."

"It would be impossible for me to put down on paper all that went on during the hours spent in the "LSD-room". Suffice it to simply say that after a period of emotional upheaval during which various phases of my childhood occurred, I finally began to realize that this session was centred around the fact that I had to make a choice, a choice as to whether I was the greater power or whether there was a God which I had to recognize and accept."

"At first I tried to bargain; I was willing to recognize a greater power but I resented strongly the fact that He might interfere with the plan of life followed by the great S.H. However, I was soon to discover that there could be no bargaining and I then decided to take refuge in reservations. I thought, very well, if I must make a choice I will do so. If I could follow a certain path of the past while accepting just certain plans for the future then I would make the choice. This did not work either and it became increasingly apparent that there could be no alternative to a complete surrender, to a clean sweep of the past. I realized that I had attempted to bargain with God all my life. I can see now why I have struggled in vain all my life, refusing to accept anything but myself. Suddenly out of nowhere, came the decision, I would make the choice, I would accept and hope to be accepted by Him. I could write for years and not be able to describe that exquisite moment of accepting and being accepted. It was without a doubt the most beautiful moment of my life and as I write this I am still amazed at the exquisite feeling of release, peace of mind, and complete realization which took place at that moment."

3. *The Breaking Down of Defences.*

The last reaction of the drug to be discussed is the breaking down of the usual defence mechanisms and the inability to rationalize past and present behaviour. This is the "mind manifesting" aspect of the drug and occurs throughout the day and late into the evening. The patient can see himself as others see him, or can, with the words, of one patient, "be his own psychiatrist". Sometimes this is facilitated in a rather literal way by a body image disturbance that makes the patient feel that he leaves his body and looks at himself as if he were standing outside himself. This aspect of the drug includes what MacLean *et al* have called "a transintegrational experience" and can be fostered by the use of a mirror. In using a mirror, the patient sees himself as he is, his appropriate and inappropriate behaviour, and can form a more realistic self-concept. At this stage, pictures of family and relatives can elicit responses of emotions and feelings that the patient may not have been aware of, and in many cases the patient is able to express hostility which he had been unable to recognize or admit under normal conditions. Other pictures, used almost as a modified TAT, also at times elicit material which has been a source of strong guilt feelings. These pictures need not necessarily have any resemblance to the patient's actual situation because the distortion of perception under LSD may make the picture appear as a picture of well-known persons or scenes in the patient's life. Also psychodrama may be of value in this connection and is facilitated by the distortion of perception and time.

The hours during which the effect of the drugs are still present but are decreasing, after the transcendental experience, are important in allowing the patient to fit the experience and new knowledge into his every day life. For the therapist, it is important at this stage to talk to the patient, keep him aware of the reality of the experience, and to refute any past rationalizations which the patient tries to use to explain away what he has learned about himself.

The therapist stays with the patient for about eight hours during the day and the

next day interviews the patient to reaffirm the reality of the experience, to bring to the surface again any painful events that may have been forgotten, and to reassure the patient that what he has learned in the session was real and significant. Most patients seem overwhelmed by the amount of material that is brought up in an LSD session and they report that it takes them days or weeks to sort out what happened and to assemble the new knowledge gained during the experience.

As an illustration of the type of person who can be helped by an LSD experience, the following are a few case histories.

Case 3. F.K.

This was a 49 year old married man, who was admitted for the first time, although he had had psychiatric treatment elsewhere on several occasions, both as an in-patient and as an out-patient.

The patient was admitted in a stupor and spent the first couple of weeks in the infirmary ward. The patient appeared to be relieved to be in hospital and seemed to have a genuine desire to stop drinking, yet he had little hope that this could be achieved. His outlook on the future was quite hopeless. He had no confidence in himself or in other people. He felt, not without reason, that his wife detested him and, having no education, being unable to do heavy labour because of his physical condition and being a known alcoholic, he felt also that he could not expect to get or to hold a job. So he would lay home brooding about this while his wife was out supporting the family and sooner or later he would invariably come to one of two conclusions, either that he wanted to commit suicide or that he wanted to get drunk and forget about it all. He had definite ideas of reference and of depersonalization and he was known to be quite hypochondriacal, probably based on an unconsciouse attitude that since he could not be a man and support his family, it would be better if he had a real physical illness instead of something that nobody would recognize as anything but lack of will power. Furthermore, if he did have cancer or heart disease, he would probably die and he would not have to worry about suicide.

On the impression that the patient's behaviour, including his drinking, was secondary to a depression, he was started on a course of ECT which he began with an attitude of "whatever you say, doctor". However, while the author was away for a week, he was apparently able to convince the nursing staff that he was too sick for further treatment after he had received only three. Nevertheless, he did appear considerably improved and said that he felt much better, paying lip service at least, to an increased acceptance of the AA, increased consideration for his wife etc. He was extremely dependant on his wife but he was so in the manner of a child, rather than a husband. This patient then underwent an LSD experience, during which very intense psychotherapeutic interaction took place and he claimed to have found a new and better way of life. At the time of latest follow-up, he had remained sober for two years.

Case 4. M.M.

A 52 year old Irish sign painter, who was admitted for the fourth time. At the time of admission he was intoxicated and rather boastful in relating drinking stories and obviously unmotivated for sobriety, although well-motivated to remaining in hospital. He received an LSD experience after which "it was felt by everybody on the ward that the patient was remarkably changed in personality and attitude" and he has now remained sober for a year. This was a patient who had spent several years on Skid Row and was well-known there. Indeed many patients later commented that "if M. can sober up, anybody can".

Case 5. E.B.

A 35 year old married farmer, who was admitted for the third time and was so uncommunicative, withdrawn and affectless that he received a diagnosis of schizo-

phrenia, simple type, with secondary chronic alcoholism. He received an LSD experience, was discharged, but began to drink again immediately and was readmitted within two weeks of his discharge. He then received a second LSD experience with 400 gamma, since the first time there had been very little reaction to 200 gamma. He now had a good reaction and has been sober for a year since this last discharge.

Results and Conclusions

Most of the patients treated have been referred by the Alcoholic Counselling Centre in Regina and since other treatment facilities are available in the Province, their policy has been to refer their most severe or chronic cases to the Saskatchewan Hospital. As a result, the material treated consisted of patients who would be considered to have a very serious prognosis by all ordinary means of evaluation. For example 87% had tried and failed with Alcoholics Anonymous at least once, 24% were separated or divorced and 57% had received some type of psychiatric treatment at least once before. Their age varied from 24 to 65, most being between 30 and 45 and the mean age being 39.3.

Of seventy patients receiving the full programme of treatment including LSD therapy and followed up between six and 18 months after discharge, 39 of 63%, had remained dry continuously since discharge or had been dry apart from a short "testing" bout of drinking immediately after discharge. Another seven, or 11%, were considered improved, that is were drinking, but definitely less than before. Sixteen, or 26%, were unimproved and eight were lost to follow-up.

Of 55 controls who were patients admitted to the hospital at a time when no beds were available to the group, and who received individual treatment by other psychiatrists, eight, or 27%, were dry, four, or 14% improved, 17, or 59%, unimproved and 26 lost to follow-up. A chi-square test showed that significantly more of the alcoholics treated with LSD were dry or improved at the time of follow-up, than patients receiving group therapy alone or of the controls ($P > .01$).

It must be stressed that although only the LSD therapy has been reported in this paper, the total two months treatment contributes to the success achieved. We also strongly agree with MacLean *et al* that LSD is not a medication in the usual sense and it is not the effect of the drug alone which is beneficial but the reaction brought about by the drug, in combination with the particular type of psychotherapeutic technique (5). LSD is not a cure for alcoholism but the total therapeutic situation can resolve some of the more basic underlying conflicts which cause an alcoholic to drink.

Summary

In this paper we have described some aspects of the LSD-25 treatment given to alcoholics in Weyburn, Saskatchewan. The three main therapeutic reactions discussed are (1) abreaction, (2) the transcendental experience and (3) break-down of inappropriate defence mechanisms. Some case histories are given to illustrate these therapeutic reactions and to give an idea of the kind of person who can be helped by this type of therapy. Some results of the work done at Weyburn are quoted and it is considered that they are quite encouraging. It is stressed that although only the LSD-25 therapy is discussed here, the total therapeutic situation contributed to the results.

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Résumé

Dans cet article, nous avons décrit certains aspects du traitement au L.S.D. 25 administré à des alcooliques, à Weyburn, Saskatchewan. Les trois principales réactions discutées sont (a) l'abréaction; (b) l'expérience transcendante et (c) l'effondrement de mécanismes de défense inappropriés. Certaines anamnèses viennent illustrer ces réactions au traitement et donnent une idée du genre de personnes que l'on peut aider au moyen de ce type de thérapie. L'article cite quelques résultats du travail exécuté à Weyburn, résultats que l'on estime encourageants. On souligne cependant que, même si l'on ne traite ici que de la thérapie par le L.S.D. 25, la situation thérapeutique globale a contribué à fournir ces résultats.